

ID:

Name:

**Brac University**

Semester: Spring 2023

Course Code: CSE250

Circuits And Electronics

Section: 05 & 06

Faculty: SHS

Set

A

Assessment: Quiz 4

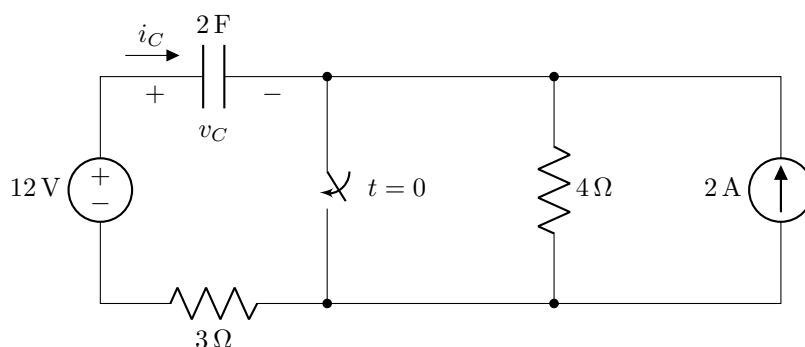
Duration: 30 minutes

Date: April 19, 2023

Full Marks (incl. bonus 4): 20

- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, both the **answer script** and the **question paper** must be returned to invigilator.
- ✓ All **1 questions** are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Bonus questions are indicated as “(bonus)” along with allotted marks.
- ✓ Write your answers inside the indicated boxes (where applicable). If you run out of room, continue on the back page.
- ✓ Symbols have their usual meanings.

■ Question 1 of 1 [CO4] [20 marks + 4 marks (bonus)]



- (a) [4 marks] Draw the initial circuit (before closing the switch) and find the initial voltage $v_C(0)$ across the capacitor. (with appropriate $\pm$ sign and unit)

$$v_C(0) = 4 \text{ V}$$

- (b) [4 marks] Draw the final circuit (after closing the switch) and find the final voltage $v_C(\infty)$ across the capacitor. (with appropriate $\pm$ sign and unit)

$$v_C(\infty) = 12 \text{ V}$$

- (c) [2 marks] Find the value of the time constant. (with appropriate $\pm$ sign and unit)

$$\tau = 6 \text{ s}$$

- (d) [4 marks] Express the capacitor voltage $v_C(t)$ as a function of time. (P.S. This should be a peicewise function, one part for $t < 0$ and other part for $t \geq 0$)

$$v_C(t) = \begin{cases} 4 & t < 0 \\ 12 - 8e^{-t/6} & t \geq 0 \end{cases}$$

- (e) [2 marks] From part (d), express the capacitor current $i_C(t)$ as a function of time. (P.S. This should be a peicewise function, one part for $t < 0$ and other part for $t \geq 0$)

$$i_C(t) = \begin{cases} 0 & t < 0 \\ 2.67e^{-t/6} & t \geq 0 \end{cases}$$

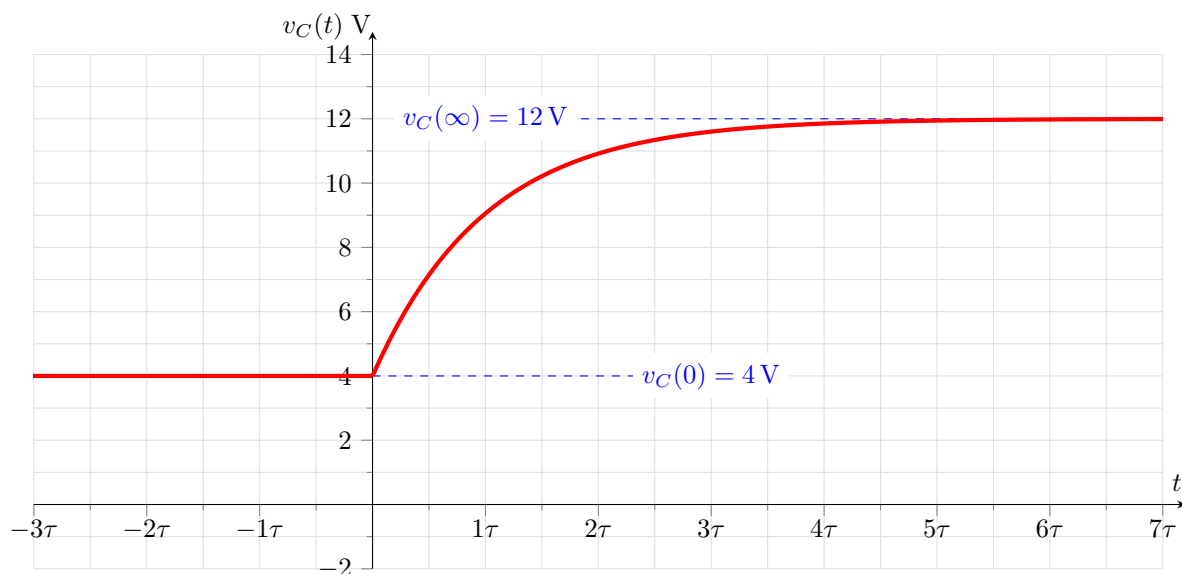
- (f) [2 marks] What's the capacitor voltage at time $t = 30 \text{ s}$? (with appropriate $\pm$ sign and unit)

$$v_C(30) = 11.95 \text{ V}$$

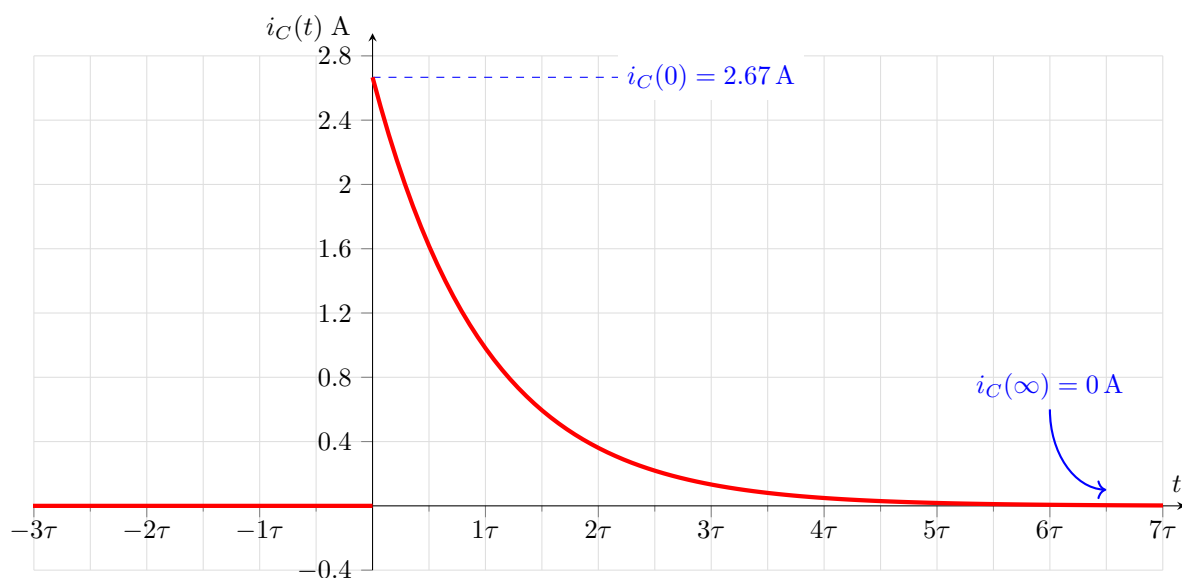
- (g) [2 marks] What's the capacitor current at time $t = 30 \text{ s}$? (with appropriate $\pm$ sign and unit)

$$i_C(30) = 0.018 \text{ A}$$

(h) [2 marks (bonus)] Plot the graph of $v_C(t)$ vs t .



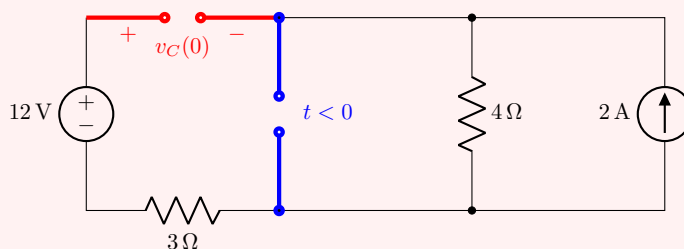
(i) [2 marks (bonus)] Plot the graph of $i_C(t)$ vs t .



Solution:

Step 1: Initial Circuit

Since the switch is being closed, initially the switch was open (blue part). Also, the capacitor is replaced with open circuit (red part). The voltage across the capacitor (open circuit) in this circuit, is the initial voltage $v_C(0)$.

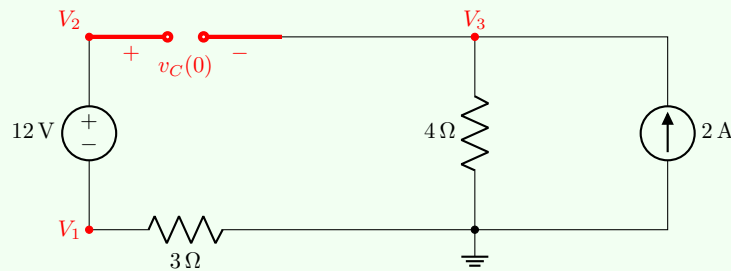


In order to find the initial voltage $v_C(0)$, we may use **Nodal Analysis**.

Step 1: Node Identification

There are **3 nodes** other than the ground node in this circuit. Here, $v_C(0) = V_2 - V_3$.

P.S. There are much easier ways to do find $v_C(0)$ if you are clever.



Step 2: Voltage Source Equation

Here we have a single independent voltage source. So we get,

$$V_2 - V_1 = 12 \quad (1)$$

Step 3: KCL at Nodes

- Since Node 1 has a current source connected to it, we can't write KCL equation for that.
- Since Node 2 has a current source connected to it, we can't write KCL equation for that.
- **Node 3** doesn't have any voltage source connected to it, so we can write KCL equation there.

$$\begin{aligned} V_3 \left(\frac{1}{4} \right) - \frac{0}{4} - 2 &= 0 \\ \Rightarrow V_3 &= 8 \end{aligned} \quad (2)$$

Step 4: KCL at Supernodes

Since we have 3 nodes and we only have 2 equations, we need a supernode equation. We have a **Floating Voltage Source** between Node 1 and 2. So we need to write a KCL equation at **Supernode (1,2)**.

$$\begin{aligned} V_1 \left(\frac{1}{3} \right) - \frac{0}{3} + V_2(0) &= 0 \\ \Rightarrow V_1 &= 0 \end{aligned} \quad (3)$$

Step 5: Solve Equations

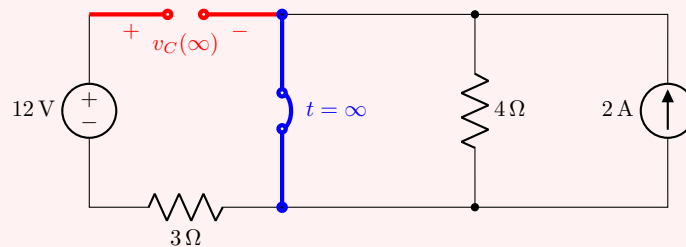
Solving the equations, we get, $V_1 = 0 \text{ V}$, $V_2 = 12 \text{ V}$, $V_3 = 8 \text{ V}$

From the nodal analysis circuit, we conclude, $v_C(0) = V_2 - V_3 = 4 \text{ V}$. So, we get,

$$v_C(0) = 4 \text{ V}$$

Step 2: Final Circuit

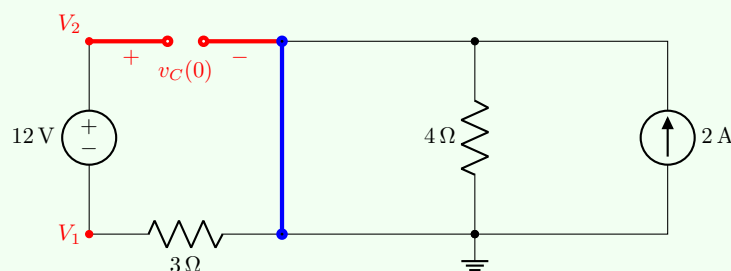
Since the switch has been closed at $t = 0$, the switch is closed (blue part) in the final circuit. Also, the capacitor is replaced with open circuit (red part). Voltage across the capacitor (open circuit) here, is the final voltage $v_C(\infty)$.



In order to find the final current $v_C(\infty)$, we can use **Nodal Analysis**.

Step 1: Node Identification

There are **2 nodes** other than the ground node in this circuit. Here, $v_C(\infty) = V_2 - 0 = V_2$.



Step 2: Voltage Source Equation

Here we have a single independent voltage source. So we get,

$$V_2 - V_1 = 12 \quad (1)$$

Step 3: KCL at Nodes

- Since Node 1 has a current source connected to it, we can't write KCL equation for that.
- Since Node 2 has a current source connected to it, we can't write KCL equation for that.

Step 4: KCL at Supernodes

Since we have 2 nodes and we only have 1 equation, we need a supernode equation. We have a **Floating Voltage Source** between Node 1 and 2. So we need to write a KCL equation at **Supernode (1,2)**.

$$\begin{aligned} V_1 \left(\frac{1}{3} \right) - \frac{0}{3} + V_2(0) &= 0 \\ \Rightarrow V_1 &= 0 \end{aligned} \quad (2)$$

Step 5: Solve Equations

Solving the equations, we get, $V_1 = 0 \text{ V}$, $V_2 = 12 \text{ V}$

From the nodal analysis circuit, we conclude, $v_C(\infty) = V_2 = 12 \text{ V}$. So, we get,

$$v_C(\infty) = 12 \text{ V}$$

Step 3: Find Time Constant τ

We know that, for capacitor, $\tau = R_{Th}C$, where C is the capacitance and R_{Th} is Thevenin's resistance across the terminals of the capacitor.

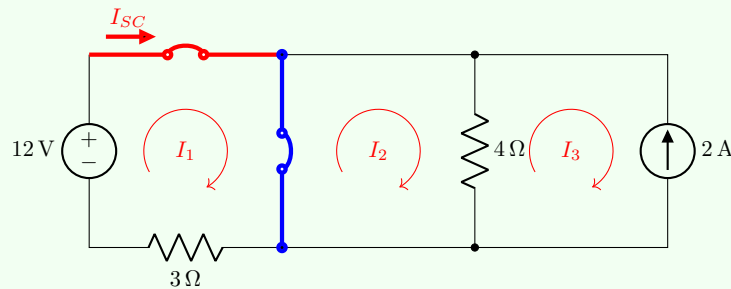
To find R_{Th} , we need V_{OC} and I_{SC} . Meaning, we need to open the capacitor to find the V_{OC} (open circuit voltage) and then short the capacitor to find the I_{SC} (short circuit current). We have actually already found V_{OC} in "Step 2: Final Circuit", because $V_{OC} = v_C(\infty) = 12\text{ V}$.

$$V_{OC} = 12\text{ V}$$

So, we only need to short the capacitor and find I_{SC} . Let's use **Mesh Analysis** to find it.

Step 1: Mesh Identification

We have **3 meshes**. Here, $I_{SC} = I_1$.



Step 2: Current Source Equation

We only have a single current source in this circuit. So we get,

$$I_3 = -2 \quad (1)$$

Step 3: KVL at Meshes

- Since Mesh 1 has no current source connected to it, we can write KVL equations here.

$$\begin{aligned} I_1(3) - 12 &= 0 \\ \Rightarrow I_1 &= \frac{12}{3} = 4 \end{aligned} \quad (2)$$

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We can stop here, since we already figured out the value of I_1 .

From the mesh analysis circuit, we get, $I_{SC} = I_1 = 4\text{ A}$.

$$I_{SC} = 4\text{ A}$$

So, we have, $R_{Th} = \frac{V_{OC}}{I_{SC}} = \frac{12\text{ V}}{4\text{ A}} = 3\Omega$.

$$R_{Th} = 3\Omega$$

Now, since, $\tau = R_{Th}C = 3\Omega \times 2\text{ F} = 6\text{ s}$.

$$\tau = 6\text{ s}$$

Step 4: Transient Equation

We know that, during transition period ($t \geq 0$), voltage across a capacitor will follow the equation,

$$v_C(t) = [v_C(0) - v_C(\infty)] e^{-t/\tau} + v_C(\infty)$$

Putting in the values from earlier steps, we get,

$$\begin{aligned} v_C(t) &= (4 - 12)e^{-t/6} + 12 \\ &= 12 - 8e^{-t/6} \end{aligned} \quad (\text{when, } t \geq 0)$$

But before switching ($t < 0$), the capacitor voltage was $v_C(0) = 4 \text{ V}$.

Keeping everything together, we can write,

$$v_C(t) = \begin{cases} 4 & t < 0 \\ 12 - 8e^{-t/6} & t \geq 0 \end{cases}$$

Step 5: Find $i_C(t)$

We can find $i_C(t)$ from $v_C(t)$ using the characteristic equation of a capacitor. After switching ($t \geq 0$),

$$\begin{aligned} i_C(t) &= C \frac{d}{dt} v_C \\ &= 2 \times \frac{d}{dt} (12 - 8e^{-t/6}) \\ &= 2(-8) \left(\frac{-1}{6} \right) e^{-t/6} \\ &= \frac{8}{3} e^{-t/6} \text{ A} = 2.67 e^{-t/6} \text{ A} \end{aligned}$$

Before switching ($t < 0$),

$$\begin{aligned} i_C(t) &= C \frac{d}{dt} v_C \\ &= 2 \times \frac{d}{dt} (4) = 2 \times 0 \\ &= 0 \text{ A} \end{aligned}$$

Keeping everything together, we can write,

$$i_C(t) = \begin{cases} 0 & t < 0 \\ 2.67 e^{-t/6} & t \geq 0 \end{cases}$$

Voltage/Current at Specific Time

If we want to find $v_C(30)$, i.e. capacitor voltage at $t = 30$, we just need to input the value of t in the equation.

$$v_C(30) = 12 - 8e^{-30/6} = 11.95 \text{ V}$$

Plotting Graphs

To plot graphs, plot each piece separately. The constant part is pretty easy to plot. Just draw a horizontal line through appropriate value.

The exponential part can be plotted by figuring out the initial and final value, and then adding them up such that it almost reaches final value after 5τ .